Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A line which makes an acute angle θ with the positive direction of x-axis is drawn through the point $P\,(3,4)$ to meet the lines $x=6$ and $y=8$ at R and S respectively then $RS=8-2\sqrt{3}$, if

(A) $\theta=\pi/3$

(B) $\theta=\pi/4$

(C) $\pi/6$

(D) $\pi/12$

Q.2

A straight line L through the point $(3,-2)$ is inclined at an angle of 60° to the line $\sqrt{3}x+y=1$. If L also intersects x-axis, then equation of L is:

(A) $y+\sqrt{3}x+2-3\sqrt{3}=0$

(B) $y-\sqrt{3}x+2+3\sqrt{3}=0$

(C) $\sqrt{3}y-x+3+2\sqrt{3}=0$

(D) $\sqrt{3}y+x-3+2\sqrt{3}=0$

Q.3

The point $(2,1)$ is translated parallel to the line $L:x-y-4=0$ by $2\sqrt{3}$ units. If the new point Q lies in the third quadrant, then the equation of the line passing through Q and perpendicular to L is:

(A) $x+y=2-\sqrt{6}$

(B) $2x+2y=1-\sqrt{6}$

(C) $x+y=3-3\sqrt{6}$

(D) $x+y=3-2\sqrt{6}$

Q.4

The area of the triangle formed by the points $(k,2-2k)$, $(-k+1,2k)$ and $(-4-k,6-2k)$ is 70 units. For

(A) four real values of k

(B) no integral value of k

@IITJEE_Advanced

(C) two integral values of k

(D) only one integral value of k

Q.5

If the vertices of a triangle ABC are $A(-4,-1)$, $B(1,2)$ and $C(4,-3)$, then the coordinates of the circumcentre of the triangle are,

(A) $(1/3,-2/3)$

(B) $(0,-4)$

(C) $(0,-2)$

(D) $(-3/2,1/2)$

Q.6

If the image of point $P(2,3)$ in a line L is $Q(4,5)$ then the image of point $R(0,0)$ in the same line is:

(A) $(2,2)$

(B) $(4,5)$

(C) $(3,4)$

(D) $(7,7)$

Q.7

Locus of the image of the point $(2,3)$ in the line $(2x-3y+4)+k(x-2y+3)=0, k \in \mathbf{R}$, is a

(A) straight line parallel to x-axis

(B) straight line parallel to y-axis

(C) circle of radius $\sqrt{2}$

(D) circle of radius $\sqrt{3}$

Q.8

If one of the lines of $my^2+\left(1-m^2\right)xy-mx^2=0$ is a bisector of the angle between the line $xy=0$ then m is

(A) $-1/2$

(B) -2

(C) 1

(D) 2

@IITJEE_Advanced

Q.9

The number of points having both coordinates as integer, that lie in the interior of the triangle with vertices $(0, 0), (0, 41)$ and $(41, 0)$ is

(A) 901

(B) 861

(C) 820

(D) 780

Q.10

The equation to the straight line passing through the point $\left(a\cos^3\theta, a\sin^3\theta\right)$ and perpendicular to the line $x\sec\theta + y\,\text{cosec}\,\theta = a$ is

(A) $x\cos\theta - y\sin\theta = a\cos 2\theta$

(B) $x\cos\theta + y\sin\theta = a\cos 2\theta$

(C) $x\sin\theta + y\cos\theta = a\cos 2\theta$

(D) none of these

Q.11

The line $3x + 2y = 24$ meets the y-axis at A and the x-axis at B; C is a point on the perpendicular bisector of AB such that the area of the triangle ABC is 91 sq. units. The coordinates of C can be

(A) $(29/2, -1)$

(B) $(29/2, 13)$

(C) $(-13/2, 1)$

(D) $(-13/2, 13)$

Q.12

Let $P = (-1, 0), Q = (0, 0)$ and $R = (3, 3\sqrt{3})$ be three points. The equation of the bisector of the $\angle PQR$ is

(A) $\sqrt{3}x + y = 0$

(B) $x + \sqrt{3}/2y = 0$

(C) $\sqrt{3}/2x + y = 0$

(D) $x + \sqrt{3}y = 0$

Q.13

@IITJEE_Advanced

Locus of centroid of the triangle whose vertices are $(a\cos t, a\sin t), (b\sin t, -b\cos t)$ and $(1, 0)$, where t is a parameter is

(A) $(3x-1)^2+(3y)^2=a^2+b^2$

(B) $(3x+1)^2+(3y)^2=a^2+b^2$

(C) $(3x+1)^2+(3y)^2=a^2-b^2$

(D) $(3x-1)^2+(3y)^2=a^2-b^2$

Q.14

Let PS be the median of the triangle with vertices $P(2, 2), Q(6, -1)$ and $R(7, 3)$. The equation of the line passing through $(1, -1)$ and parallel to PS is

(A) $2x-9y-7=0$

(B) $2x-9y-11=0$

(C) $2x+9y-11=0$

(D) $2x+9y+7=0$

Q.15

If the vertices A and B of a triangle ABC are given by $(2, 5)$ and $(4, -11)$ and C moves along the line $L_1: 9x+7y+4=0$, the locus of the centroid of the triangle ABC is a straight line parallel to

(A) AB

(B) BC

(C) CA

(D) L_1

Q.16

Image of the point $(-1, 3)$ with respect to the line $y=2x$ is

(A) $(7/5, 14/5)$

(B) $(1, 2)$

(C) $(3, 1)$

(D) $(5, 1)$

Q.17

A ray of light is incident along a line which meets another line $7x-y+1=0$, at the point $(0, 1)$. The ray is then reflected from this point along the line, $y+2x=1$. Then the equation of the line of incidence of the ray of light is:

@IITJEE_Advanced

(A) $41x - 25y + 25 = 0$

(B) $41x + 25y - 25 = 0$

(C) $41x - 38y + 38 = 0$

(D) $41x + 38y - 38 = 0$

Q.18

The locus represented by the equation $(x - y + c)^2 + (x + y - c)^2 = 0$ is

(A) a line parallel to x-axis

(B) a point

(C) a pair of straight lines

(D) a line parallel of y-axis

Q.19

If $y = m_i x + \frac{1}{m_i}(i = 1, 2, 3)$ represents three straight lines whose slopes are the roots of the equation. $2m^3 - 3m^2 - 3m + 2 = 0$, A and B are the algebraic sum of the intercepts made by the lines on x-axis and y-axis respectively, then $\alpha A + \beta B = 0$ if (α, β) is

(A) $(4, 7)$

(B) $(2, 7)$

(C) $(7, 2)$

(D) $(-1, -7)$

Q.20

The centroid of a triangle lies at the origin and the coordinates of its two vertices are $(-8, 7)$ and $(9, 4)$. The area of the triangle is

(A) $95/6$

(B) $285/2$

(C) $190/3$

(D) 285

Q.21

A straight line L with negative slope passes through the point $(8, 2)$ and cuts the positive coordinate axes at points P and Q. The absolute minimum value of $OP + OQ$ as L varies, where O is the origin, is

@IITJEE_Advanced

Q.22

The absolute value of the sum of the abscissas of all the points on the line $x + y = 4$ that lie at a unit distance from the line $4x + 3y - 10 = 0$ is

Q.23

For all real values of a and b, lines $(2a + b)x + (a + 3b)y + (b - 3a) = 0$ and $mx + 2y + 6 = 0$ are concurrent. Then m is equal to (mark answer in negative value)

Q.24

If P is a point (x, y) on the line. $\mathrm{y} = -3\mathrm{x}$ such that P and the point $(3, 4)$ are on the opposite sides of the line $3x - 4y = 8$ then x > a and y < b, where a is maximum possible value. Find 15a.

Q.25

Two sides of a rectangle are $3x + 4y + 5 = 0, 4x - 3y + 15 = 0$ and one of its vertices is $(0, 0)$. The area of rectangle is

Q.26

Triangle ABC with $AB = 13, BC = 5$, and $AC = 12$ slides on the coordinates axes with A and B on the positive x-axis and positive y-axis respectively. The locus of vertex C is a line $12x - ky = 0$. Then the value of k is

Q.27

The slopes of the line which passes through the origin, and the mid-point of the line segment joining the points. $P(3, -4)$ and $Q(-5, -2)$ is

Q.28

Let $A_0A_1A_2A_3A_4A_5$ be a regular hexagon inscribed in a circle of unit radius then the product of the lengths of the line segments A_0A_1, A_0A_2 and A_0A_4 is

Q.29

The value of p for which the lines $2x + y - 3 = 0$ $3x - y - 2 = 0$ and $x - py + 5 = 0$ may intersect at a point is

Q.30

@IITJEE_Advanced

If the line $2x + y = k$ passes through the point which divides the segment joining the points $(1, 1)$ and $(2, 4)$ in the ratio $3 : 2$, then k equals

Answers & Solutions

Q.1 Answer:

$\pi/6$

Solution:

$$\frac{x-3}{\cos\theta} = \frac{y-4}{\sin\theta} = r$$

$$PR = 3|\cos\theta, PS = 4|\sin\theta$$

$$RS = PS - PR$$

$$\Rightarrow 8 - 2\sqrt{3} = \frac{4}{\sin\theta} - \frac{3}{\cos\theta}$$

$$\Rightarrow 2 - \frac{\sqrt{3}}{2} = \frac{1}{\sin\theta} - \frac{3}{4\cos\theta}$$

We can observe that

$\sin\theta = \frac{1}{2}$ and $\cos\theta = \frac{\sqrt{3}}{2}$

which Implies, $\theta = \frac{\pi}{6}$

Q.2 Answer:

$y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$

Solution:

Given line is

$\sqrt{3}x + y = 1$

Slope of this line is $m_2 = -\sqrt{3}$

Let the slope of line L **be** m_1**.**

Now, angle between the lines is 60°

$$\tan 60^\circ = \left|\frac{m_1 + \sqrt{3}}{1 - \sqrt{3}\, m_1}\right|$$

$$\Rightarrow \pm\sqrt{3} = \frac{m_1 + \sqrt{3}}{1 - \sqrt{3}m_1}$$

$$\Rightarrow m_1 = 0, m_1 = \sqrt{3}$$

Equation of line L passing through $(3, -2)$ **is**

$$y + 2 = \sqrt{3}(x - 3)$$

$$\Rightarrow y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$$

Q.3 Answer:

$x + y = 3 - 2\sqrt{6}$

@IITJEE_Advanced

Solution:

An equation of line through $(2,1)$ and parallel to $x-y=4$ is $(x-2)-(y-1)=0$ or $x-y-1=0$

A point on this line is $(t, t-1)$.

Its distance from $(2,1)$ is

$$d=\sqrt{(t-2)^2+(t-1-1)^2}=\sqrt{2}|t-2|$$

Now $\sqrt{2}|t-2|=2\sqrt{3}$

$$\Rightarrow t-2=\pm\sqrt{6} \Rightarrow t=2\pm\sqrt{6}$$

As the point lies in the third quadrant, we take $t=2-\sqrt{6}$

$$\therefore \text{ point } Q \text{ is } (2-\sqrt{6}, 1-\sqrt{6})$$

An equation of required line through Q and perpendicular to L is

$$(x-(2-\sqrt{6}))+(y-(1-\sqrt{6}))=0$$

or $x+y=3-2\sqrt{6}$

Q.4 Answer:

only one integral value of k

Solution:

$$\begin{vmatrix} k & 2-2k & 1 \\ -k+1 & 2k & 1 \\ -4-k & 6-2k & 1 \end{vmatrix} = 140$$

which is true for only one integral value of k.

Q.5 Answer:

$(0,-2)$

Solution:

Given,

Vertices of $\triangle ABC$ are

@IITJEE_Advanced

$A(-4,-1), B(1,2), c(4,-3)$

From the distance formula

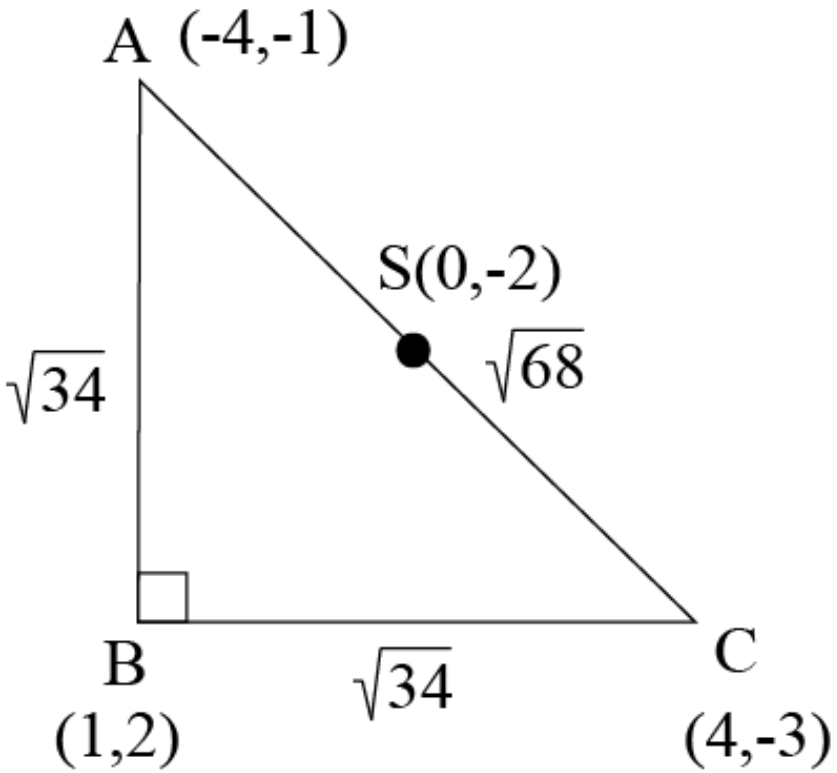
i.e, Distance between points (x_1, y_1) , (x_2, y_2)

$$\text{is } \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$\Rightarrow AB = \sqrt{5^2+3^2} = \sqrt{34}$$
$$BC = \sqrt{3^2+5^2} = \sqrt{34}$$
$$CA = \sqrt{8^2+2^2} = \sqrt{68}$$

we can observe that

$$AB^2 + BC^2 = CA^2$$



$\Rightarrow$ **Triangle is right angled at** B**, So mid-point of** AC **is the circumcentre.**

$\Rightarrow$ **Circumcentre(s)** $= (0,-2)$.

Q.6 Answer:

$(7,7)$

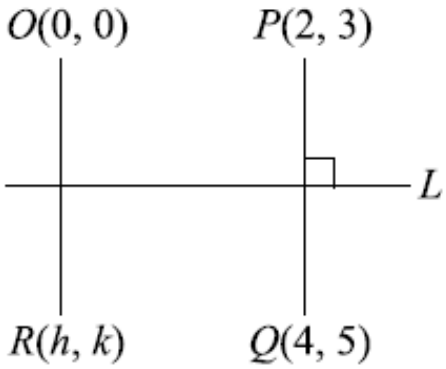
Solution:

@IITJEE_Advanced

Slope of $PQ = 1$

Mid point of PQ is $(3, 4)$. So equation of L is

$$y - 4 = -(x - 3)$$
$$\Rightarrow \quad x + y - 7 = 0$$



Let the image of $\theta(0, 0)$ be $R(h, k)$ the slope of $OR =$ slope of PQ

$$\Rightarrow \quad \frac{k}{h} = 1 \quad \Rightarrow \quad k = h$$

Also mid-point $\left(\frac{h}{2}, \frac{k}{2}\right)$ of OR lies on $L \therefore$ $\quad \frac{h}{2} + \frac{k}{2} - 7 = 0 \quad \Rightarrow \quad h = k = 7$

and the required point is $(7, 7)$

Q.7 Answer:

circle of radius $\sqrt{2}$

Solution:

$(2x - 3y + 4) + k(x - 2y + 3) = 0, k \in R$, represents a family of lines passing through the intersection of the lines $2x - 3y + 4 = 0$ and $x - 2y + 3 = 0$, that is, through $(1, 2)$

If (x, y) is image of $(2, 3)$ in (1), then both (x, y) and $(2, 3)$ are equidistant from $(1, 2)$, thus,

$$(x - 1)^2 + (y - 2)^2 = (2 - 1)^2 + (3 - 2)^2 = 2$$

Thus, (x, y) lies on a circle of radius $\sqrt{2}$

Q.8 Answer:

@IITJEE_Advanced

Solution:

The bisectors of $xy = 0$ ie $x = 0$ and $y = 0$, we have are
$x + y = 0$ and $x - y = 0$
$\Rightarrow$ Now
$my^2 + \left(1 - m^2\right) xy - mx^2 = 0$
$my^2 + 1xy - m^2xy - mx^2 = 0$
$my(my + x) - mx(my + x) = 0$
$\Rightarrow (y - mx)(my + x) = 0$
on comparing there with $x + y = 0$ and $x - y = 0$ we have
$m = 1$

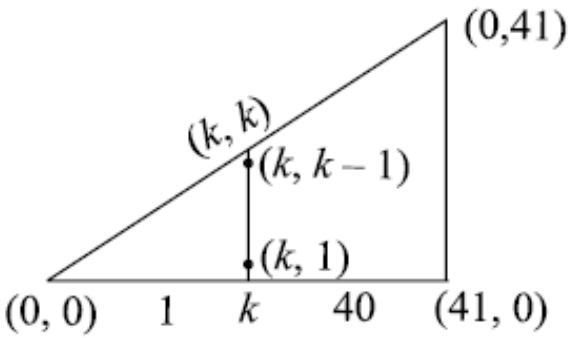
Q.9 Answer:

780

Solution:

For $1 \leq k \leq 40$, the number of integral points on the line $x = k$ and lying in the interior of triangles is $k - 1$

Thus, required number of points $= \sum_{k=1}^{40}(k-1)\ \frac{1}{2}(39)(40) = 780$



Q.10 Answer:

$x\cos\theta - y\sin\theta = a\cos 2\theta$

Solution:

Equation of perpendicular on line

$$x\sec\theta + y\csc\theta = a$$
$$x\csc\theta - y\sec\theta = \lambda$$

This line passes through point $\left(a\cos^3\theta, a\sin^3\theta\right)$, then

$$\frac{a\cos^3\theta}{\sin\theta} - \frac{a\sin^3\theta}{\cos\theta} = \lambda$$

$$\frac{a\left(\cos^4\theta - \sin^4\theta\right)}{\sin\theta\cos\theta} = \lambda$$

@IITJEE_Advanced

$$\frac{a\left(\cos^2\theta-\sin^2\theta\right)\left(\cos^2\theta+\sin^2\theta\right)}{\sin\theta\cos\theta}=\lambda$$

$$\frac{a\cos 2\theta}{\sin\theta\cos\theta}=\lambda$$

Putting the value of λ in Eq. (i)

$$x\csc\theta-y\sec\theta=\frac{a\cos 2\theta}{\sin\theta\cos\theta}$$

$$x\cos\theta-y\sin\theta=a\cos 2\theta$$

Q.11 Answer:

$(29/2, 13)$

Solution:

$A(0,12), B(8,0), C(x,y)$ **lies on the line**

$$2x-3y+10=0 \text{ and } \frac{1}{2}\begin{vmatrix} x & y & 1 \\ 0 & 12 & 1 \\ 8 & 0 & 1 \end{vmatrix}=\pm 91$$

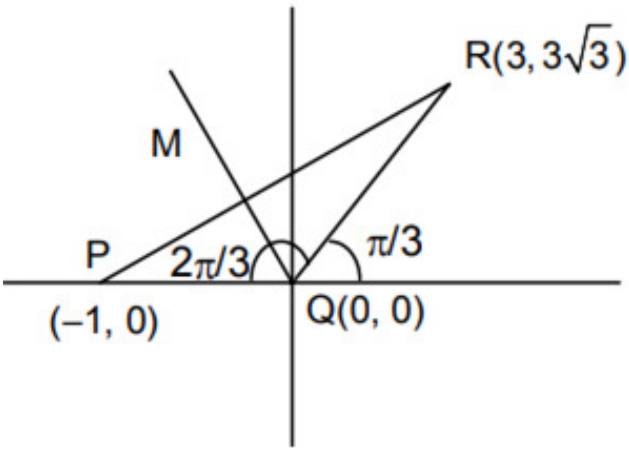
Solving we get $(x,y)=(-13/2,-1)$ **or** $(29/2,13)$

Q.12 Answer:

$\sqrt{3}x+y=0$

Solution:

Now, slope of $QR=\frac{3\sqrt{3}-0}{3-0}=\sqrt{3}=\tan\theta$



$\Rightarrow \quad \theta=\frac{\pi}{3}$

The angle between PQR **is** $\frac{2\pi}{3}$**, so the line** QM **makes an angle** $\frac{2\pi}{3}$ **from positive direction of** X**-axis.**

@IITJEE_Advanced

Slope of the line $QM = \tan\frac{2\pi}{3} = -\sqrt{3}$
Hence, equation of line QM **is** $y = -\sqrt{3}x$
or $\sqrt{3}x + y = 0$

Q.13 Answer:

$(3x-1)^2 + (3y)^2 = a^2 + b^2$

Solution:

Let (h, k) **be the centroid of the triangle, then**

$$h = \frac{a\cos t + b\sin t + 1}{3}, k = \frac{a\sin t - b\cos t}{3}$$
$$\Rightarrow (3h-1)^2 + (3k)^2 = a^2 + b^2$$

Locus of (h, k) **is**

$$(3x-1)^2 + (3y)^2 = a^2 + b^2$$

Q.14 Answer:

$2x + 9y + 7 = 0$

Solution:

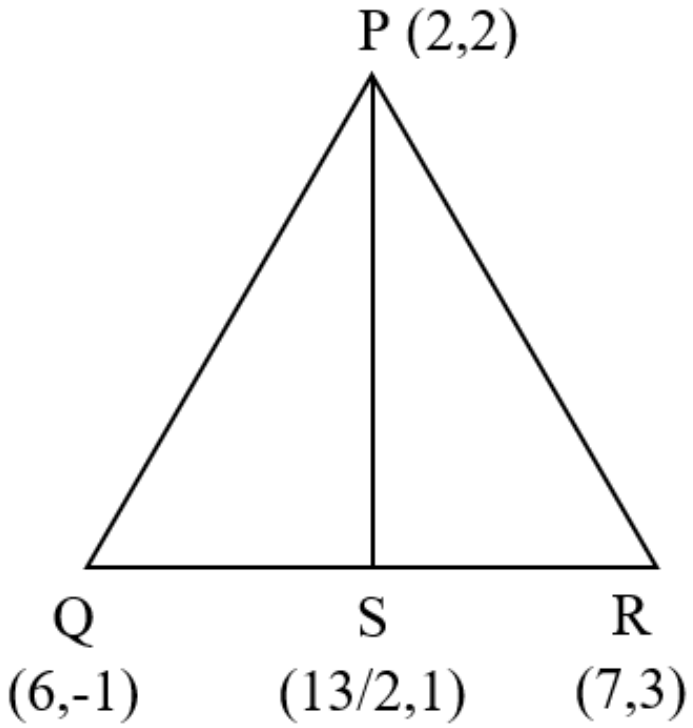
Given,

Vertices of the triangle.

$P(2, 2), Q(6, -1), R(7, 3)$

Let PS be median of triangle

@IITJEE_Advanced



$S = \frac{Q+R}{2}$

$'S'$ is mid-point of $Q \& R$

$\Rightarrow s = (13/2, 1)$

slope of line $PS = \frac{-2}{9}$

The line passing through $(1, -1)$ and slope $= \frac{-2}{9}$ is

$$y - y_1 = m(x - x_1)$$
$$y + 1 = -\frac{2}{9}(x - 1)$$
$$9y + 9 = -2x + 2$$
$$2x + 9y + 7 = 0$$

Q.15 Answer:

L_1

Solution:

Let coordinate of $\mathrm{C}(\mathrm{x}, \mathrm{y})$ which satisfies the line L.

And Coordinates of centriod (h, k),

@IITJEE_Advanced

$\mathrm{h}=\frac{2+4+\mathrm{x}}{3}$ **or,** $3\,\mathrm{h}=6+\mathrm{x}$ **or,** $x=3\,\mathrm{h}-6$

$\mathrm{k}=\frac{5-11+\mathrm{y}}{3}$ **or,** $3\mathrm{k}=-6+\mathrm{y}$ **or,** $\mathrm{y}=3\mathrm{k}+6$

(ii)

Putting value of x **and y from eqn.(i) and (ii) in line** L **we get,**

$$9(3h-6)+7(3k+6)+4=0$$

$$\text{or, } 27\,\mathrm{h}-54+21\mathrm{k}+42+4=0$$
$$\text{or, } 27\,\mathrm{h}+21\mathrm{k}-8=0$$

Now, Locus of Centroid is $27\mathrm{x}+21\mathrm{y}-8=0$

$$\rightarrow y=\frac{8}{21}-\frac{9}{7}x$$

slope is $\mathrm{m}=-\frac{9}{7}$ **exactly as the slope of line** L **thus we can say that the locus is parallel to the line** L**.**

Q.16 Answer:

$(3,1)$

Solution:

Line is $y-2x=0$

Let the image of $(-1,3)$ **be** (h,k)**.**

Then $(\mathrm{h}-\mathrm{x}_1)/\mathrm{a}=(\mathrm{k}-\mathrm{y}_1)/\mathrm{b}=-2\left(\mathrm{ax}_1+\mathrm{by}_1+\mathrm{c}\right)/\left(\mathrm{a}^2+\mathrm{b}^2\right)$

Here $x_1=-1, y_1=3, a=-2, b=1, c=0$

$$\Rightarrow (h+1)/-2=(k-3)/1=-2(2+3+0)/\left(2^2+1^2\right)$$
$$=>-(h+1)/2=(k-3)=-2$$
$$\Rightarrow h+1=4 \text{ and } k-3=-2$$
$$=>h=3 \text{ and } k=1$$

So the image is $(3,1)$**.**

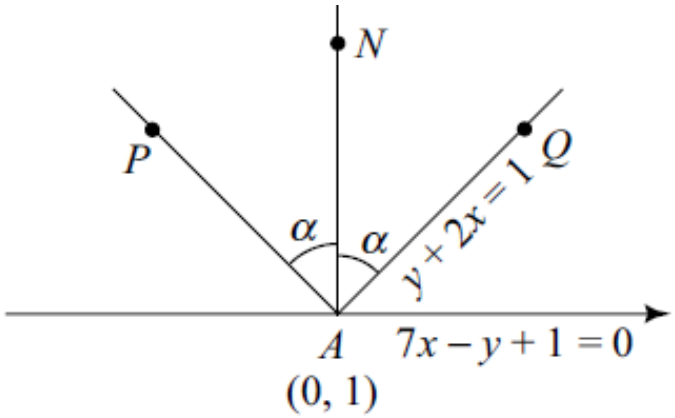
Q.17 Answer:

$41x-38y+38=0$

Solution:

@IITJEE_Advanced

An equation of normal AN is $(x-0)+7(y-1)=0$ or $x+7y-7=0$



Let an equation of PA be $y=mx+1$, then $\angle PAN=\angle NAQ$

$$\frac{-1/7-m}{1-m/7}=\frac{-2-(-1/7)}{1+(2/7)}$$
$$\Rightarrow \left(\tfrac{1}{7}+m\right)(9)=13\left(1-\tfrac{m}{7}\right)$$
$$\Rightarrow \left(9+\tfrac{13}{7}\right)m=13-\tfrac{9}{7}$$
$$\Rightarrow 76m=82 \Rightarrow m=41/38$$

Thus, required equation is

$y=\frac{41}{38}x+1$

or $41x-38y+38=0$

Q.18 Answer:

a point

Solution:

$(x-y+c)^2+(x+y-c)^2=0$

We know that the sum of the squares of two real quantities cannot be zero unless each of the squares is separately zero.

The only real points that satisfy the equation (1) therefore satisfy both of the equations

$x-y+c=0$ **and** $x+y-c=0$

But the only solution of these two equations is

@IITJEE_Advanced

$x = 0$ and $y = c$.

The only real point represented by equation (1) is therefore $(0, c)$.

The same result may be obtained in a different manner. The equation (1) gives

$$(x - y + c)^2 = -(x + y - c)^2$$
$$x - y + c = \pm\sqrt{-1}(x + y - c)$$

It therefore represents the two imaginary straight lines

$$x(1 - \sqrt{-1}) - y(1 + \sqrt{-1}) + c(1 + \sqrt{-1}) = 0$$
$$x(1 + \sqrt{-1}) - y(1 - \sqrt{-1}) + c(1 - \sqrt{-1}) = 0$$

Each of these two straight lines passes through the real point $(0, c)$. We may therefore say that (1) represents two imaginary straight lines passing through the point $(c, 0)$.

Q.19 Answer:

$(2, 7)$

Solution:

$(m + 1)(2m - 1)(m - 2) = 0$

$\Rightarrow m_1 = -1, m_2 = 1/2, m_3 = 2$

$A = -\sum \frac{1}{m_i^2} = -\frac{21}{4}$

$B = \sum \frac{1}{m_i} = \frac{3}{2}$

$-\frac{21}{4}\alpha + \frac{3}{2}\beta = 0$

$\Rightarrow -21\alpha + 6\beta = 0$

which is satisfied by (b)

Q.20 Answer:

$285/2$

Solution:

We know the centroid of a triangle is $\{(x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3\}$

Here centroid is $(0, 0)$ and two vertices are $(-8, 7)$ and $(9, 4)$

@IITJEE_Advanced

So $(x_1 - 8 + 9)/3 = 0$ **and** $(y_1 + 7 + 4) = 0$

$\Rightarrow x_1 + 1 = 0$ **and** $y_1 + 11 = 0$

$\Rightarrow x_1 = -1, y_1 = -11$

So the third vertex is $(-1, -11)$

Area of triangle $\Delta = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

$$= \frac{1}{2}\begin{vmatrix} -1 & -11 & 1 \\ -8 & 7 & 1 \\ 9 & 4 & 1 \end{vmatrix}$$
$$= |(1/2)[-1(7-4) + 11(-8-9) + 1(-32-63)]$$
$$= |(1/2)[-3 - 187 - 95]|$$
$$= |-285/2|$$
$$= 285/2$$

Q.21 Answer:

—

Solution:

equtions of line: $\frac{x}{a} + \frac{y}{b} = 1$

Since line intersect co-ordinate axis at P & Q, then $\frac{x}{OP} + \frac{y}{OQ} = 1$

and line passes through

so $\frac{8}{op} + \frac{2}{0Q} = 1$

$OP = \left(\frac{8OQ}{OQ-2}\right)$

so $OP + OQ = \frac{80Q}{OQ-2}$ **+OQ**

To find its $\min$ **value,**

We will differentiate the above equation

@IITJEE_Advanced

$\frac{d(OP+OQ)}{d(OQ)} \Rightarrow \frac{(OQ-2)-8OQ(1)+1}{(OQ-2)^2}$

$O = \frac{1}{16}1 - \frac{16}{(OQ-2)^2}$

$1 = \frac{16}{(OQ-2)^2}$

$(OQ-2) = \pm y$

$OQ = 6x - 2$

so $OQ = 6$

$$\therefore OP + OQ = \frac{8(6)}{6-2} + 6$$
$$= \frac{48}{4} + 6 = 12 + 6$$

Thos $OP + OQ = 18$

Q.22 Answer:

—

Solution:

Let $(\mathrm{h}, 4-\mathrm{h})$ **is a point on** $\mathrm{x}+\mathrm{y}=4$ **line**

So, distance of $(\mathrm{h}, 4-\mathrm{h})$ **from** $4\mathrm{x}+3\mathrm{y}-10=0$ **line is**

$$\mathrm{d} = \left|\frac{4\,\mathrm{h}+3(4-\mathrm{h})-10}{5}\right| = 1$$
$$\left|\frac{\mathrm{h}+2}{5}\right| = 1$$
$$|\,\mathrm{h}+2| = 5$$
$$\mathrm{h} = 3, -7$$

So, sum of the abscissa of all point on $\mathrm{x}+\mathrm{y}=4$ **which is unit distance from**

$$4x + 3y - 10 = 0$$
$$= 3 - 7$$
$$= -4$$

Q.23 Answer:

—

Solution:

@IITJEE_Advanced

Given equation $eq(2a+b)x+(a+3b)y+(b-3a)=0\, mx+2y+6=0$ are concurrent real ralues of a and b, so they must be

Present the same line for the same values of a and b, then

$\frac{2a+b}{m}=\frac{a+3b}{2}=\frac{b-3a}{6}$

Taking 2ast two rafio $\frac{a+3b}{2}=\frac{-3a+b}{6}$

$\Rightarrow b=\frac{-3}{4}a$

Taking first two ratio

$$\begin{aligned} m &= \frac{2(2a+b)}{a+3b} \\ &= \frac{2\left[2a-\left(\frac{3}{4}\right)\right]a}{a+3\left(\frac{-3}{4}\right)a} \\ &= \frac{10}{-5}=-2 \end{aligned}$$

Q.24 Answer:

—

Solution:

Let $\mathrm{k}=3\mathrm{x}-4\mathrm{y}-8$ then the value of k at $(3,4)=3\times 3-4\times 4-8=-15<0$ $\therefore$ For the point $P(x,y)$ we should have
$\mathrm{k}>0\Rightarrow 3\mathrm{x}-4\mathrm{y}-8>0\Rightarrow 3\mathrm{x}-4(-3\mathrm{x})-8>0$
$[\because \mathrm{P}(\mathrm{x},\mathrm{y})$ lies on $\mathrm{y}=-3\mathrm{x}]\Rightarrow;\mathrm{x}>8/15$ and $-\mathrm{y}-4\mathrm{y}-8>0\Rightarrow \mathrm{y}<-8/5.$

Q.25 Answer:

—

Solution:

As $(0,0)$ Does not lies on $3\mathrm{x}+4\mathrm{y}+5=0, 4\mathrm{x}+3\mathrm{y}+15=0$

Perpendicular distance from $(0,0)$ to $3x+4y+5=0$ is

$A=\left|\frac{3.0+4.0+5}{\sqrt{3^2+4^2}}\right|=1$

And perpendicular distance from $(0,0)$ to $4x+3y+15=0$ is

@IITJEE_Advanced

$B = \left| \frac{4.0+3.0+15}{\sqrt{4^2+3^2}} \right| = 3$

Then area as $\mathrm{AB} = 1 \times 3 = 3$ **sq. units.**

Q.26 Answer:

—

Solution:

$\tan(90^\circ - \theta) = \frac{BC}{AC}$

Substituting the values of BC **and** AC **and the value of** $\tan(90 - \theta)$ **as** $\cot\theta$ **in the above equation we get,** $\cot\theta = \frac{5}{12}$

As $\tan\theta$ **is the reciprocal of** $\cot\theta$ **so the value of** $\tan\theta$ **is** $\frac{12}{5}$**.**

Now, we have to find the value of k **in** $12x - ky = 0$**.**

We know that slope of the line is:

$$\text{slope } = -\frac{(\text{ coefficient of x})}{(\text{ coefficient of y})}$$

So, the slope of the line $12x - ky = 0$ **is equal to** $\frac{12}{k}$

And slope $= \tan\theta$**. So, we are going to equate the above slope with the value of** $\tan\theta$ **that we have solved above.**

$$\frac{12}{5} = \frac{12}{k}$$
$$\Rightarrow k = 5$$

Hence, the value of k **is equal to 5.**

Q.27 Answer:

—

Solution:

Coordinate of the mid-point of the line segment joining P **and** Q **are** $\left(\frac{3-5}{2}, \frac{-4-2}{2}\right) = (-1, -3)$ **so slope of the required line is** $\frac{0-(-3)}{0-(-1)} = 3$

Q.28 Answer:

—

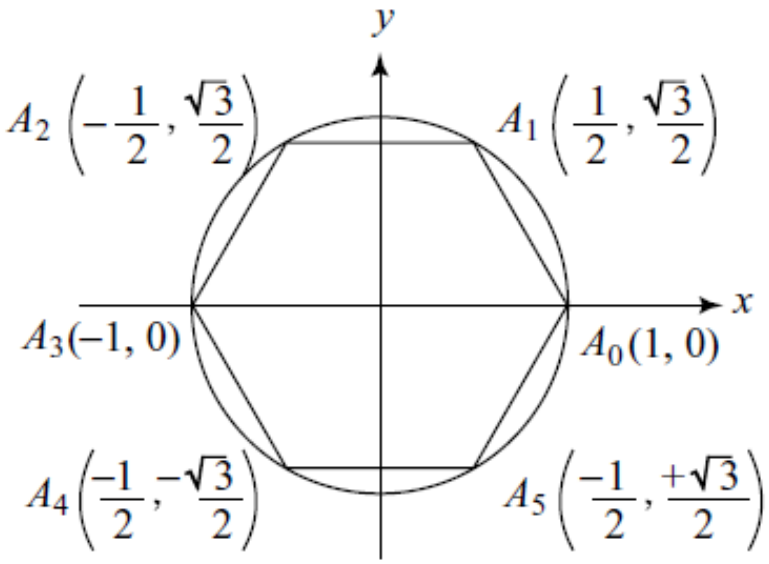
Solution:

@IITJEE_Advanced

The vertices of the hexagon be as shown in the figure

$$(A_0A_1)^2 = \frac{1}{4} + \frac{3}{4} = 1$$
$$(A_0A_2)^2 = \frac{9}{4} + \frac{3}{4} = 3$$
$$(A_0A_4)^2 = (A_0A_2)^2 = 3$$

So $(A_0A_1)(A_0A_2)(A_0A_4) = \sqrt{1 \times 3 \times 3} = 3$



Q.29 Answer:

—

Solution:

$2x + y - 3 = 0$ **.....(1)**

$3x - y - 2 = 0$ **....(2)**

$x - py + 5 = 0$ **.....(3)**

Solving (1) & (2)

$5x = 5$

$\Rightarrow x = 1, \quad y = 1$

Putting values of x & y in equation 3

$$(1) - p(1) + 5 = 0$$
$$p = 6$$

@IITJEE_Advanced

Q.30 Answer:

—

Solution:

Coordinates of the point dividing the join of the points $(1,1),(2,4)$ **in the ratio** $3:2$ **are** $\left(\frac{3\times2+2\times1}{3+2},\frac{3\times4+2\times1}{3+2}\right)=\left(\frac{8}{5},\frac{14}{5}\right)$

$2x+y=k$ **passes through this points.**

if $2\times(8/5)+(14/5)=k\Rightarrow k=6$

@IITJEE_Advanced